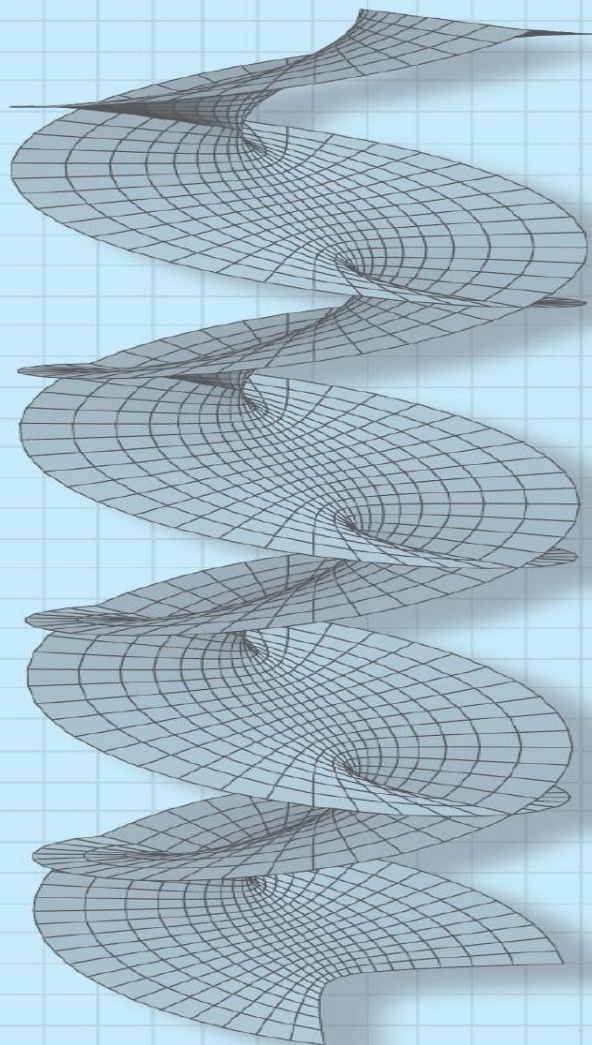




A First Course in

Complex Analysis

with Applications

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If a complex function f fails to be analytic at a point $z = z_0$, then this point is said to be a singularity or singular point of the function. For example, the complex numbers $z = 2i$ and $z = -2i$ are singularities of the function $f(z) = z/(z^2 + 4)$ because f is discontinuous at each of these points.

In this section we will be concerned with a new kind of “power series” expansion of f about an isolated singularity z_0 . This new series will involve negative as well as nonnegative integer powers of $z - z_0$.

Laurent Series

Isolated Singularities

Suppose that $z = z_0$ is a singularity of a complex function f . The point $z = z_0$ is said to be an **isolated singularity** of the function f if there exists *some* deleted neighborhood, or punctured open disk, $0 < |z - z_0| < R$ of z_0 throughout which f is analytic. For example, we have just seen that $z = 2i$ and $z = -2i$ are singularities of $f(z) = z/(z^2 + 4)$. Both $2i$ and $-2i$ are isolated singularities since f is analytic at every point in the neighborhood defined by $|z - 2i| < 1$, except at $z = 2i$, and at every point in the neighborhood defined by $|z - (-2i)| < 1$, except at $z = -2i$. In other words, f is analytic in the deleted neighborhoods $0 < |z - 2i| < 1$ and $0 < |z + 2i| < 1$.

A New Kind of Series

If $z = z_0$ is a singularity of a function f , then certainly f cannot be expanded in a power series with z_0 as its center. However, about an isolated singularity $z = z_0$, it is possible to represent f by a series involving both negative and nonnegative integer powers of $z - z_0$; that is,

$$f(z) = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots \quad (1)$$

As a *very* simple example of (1) let us consider the function $f(z) = 1/(z - 1)$. As can be seen, the point $z = 1$ is an isolated singularity of f and consequently the function cannot be expanded in a Taylor series centered at that point. Nevertheless, f can be expanded in a series of the form given in (1) that is valid for all z *near* 1:

$$f(z) = \cdots + \frac{0}{(z - 1)^2} + \frac{1}{z - 1} + 0 + 0 \cdot (z - 1) + 0 \cdot (z - 1)^2 + \cdots \quad (2)$$

The series representation in (2) is valid for $0 < |z - 1| < \infty$.

Using summation notation, we can write (1) as the sum of two series

$$f(z) = \sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z - z_0)^k. \quad (3)$$

The two series on the right-hand side in (3) are given special names. The part with negative powers of $z - z_0$, that is,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \quad (4)$$

is called the **principal part** of the series (1) and will converge for $|1/(z - z_0)| < r^*$ or equivalently for $|z - z_0| > 1/r^* = r$. The part consisting of the nonnegative powers of $z - z_0$,

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k, \quad (5)$$

is called the **analytic part** of the series (1) and will converge for $|z - z_0| < R$. Hence, the sum of (4) and (5) converges when z satisfies both $|z - z_0| > r$

and $|z - z_0| < R$, that is, when z is a point in an annular domain defined by $r < |z - z_0| < R$.

By summing over negative and nonnegative integers, (1) can be written compactly as

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k.$$

EXAMPLE 1 Series of the Form Given in (1)

The function $f(z) = \frac{\sin z}{z^4}$ is not analytic at the isolated singularity $z = 0$ and hence cannot be expanded in a Maclaurin series. However, $\sin z$ is an entire function, we know that its Maclaurin series,

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \frac{z^9}{9!} - \cdots,$$

converges for $|z| < \infty$. By dividing this power series by z^4 we obtain a series for f with negative and positive integer powers of z :

$$f(z) = \frac{\sin z}{z^4} = \overbrace{\frac{1}{z^3} - \frac{1}{3!z}}^{\text{principal part}} + \overbrace{\frac{z}{5!} - \frac{z^3}{7!} + \frac{z^5}{9!} - \cdots}^{\text{analytic part}}. \quad (6)$$

The analytic part of the series in (6) converges for $|z| < \infty$. (Verify.) The principal part is valid for $|z| > 0$. Thus (6) converges for all z except at $z = 0$; that is, the series representation is valid for $0 < |z| < \infty$.

A series representation of a function f that has the form given in (1), and (2) and (6) are such examples, is called a **Laurent series** or a **Laurent expansion** of f about z_0 on the annulus $r < |z - z_0| < R$.

Theorem Laurent's Theorem

Let f be analytic within the annular domain D defined by $r < |z - z_0| < R$. Then f has the series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k \quad (7)$$

valid for $r < |z - z_0| < R$. The coefficients a_k are given by

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(s)}{(s - z_0)^{k+1}} ds, \quad k = 0, \pm 1, \pm 2, \dots, \quad (8)$$

where C is a simple closed curve that lies entirely within D and has z_0 in its interior.

EXAMPLE 2 Four Laurent Expansions

Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for the following annular domains.

- (a) $0 < |z| < 1$ (b) $1 < |z|$ (c) $0 < |z-1| < 1$ (d) $1 < |z-1|$

Solution The four specified annular domains are shown in Figure . The black dots in each figure represent the two isolated singularities, $z = 0$ and $z = 1$, of f . In parts (a) and (b) we want to represent f in a series involving only negative and nonnegative integer powers of z , whereas in parts (c) and (d) we want to represent f in a series involving negative and nonnegative integer powers of $z-1$.

(a) By writing

$$f(z) = -\frac{1}{z} \frac{1}{1-z},$$

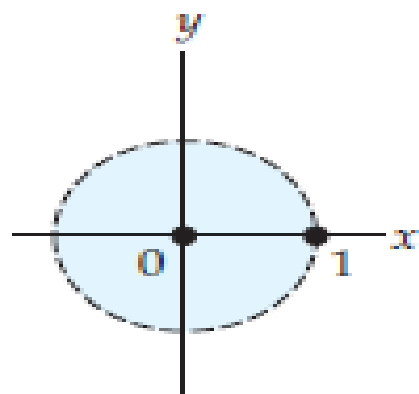
to write $1/(1-z)$ as a series:

$$f(z) = -\frac{1}{z} [1 + z + z^2 + z^3 + \cdots].$$

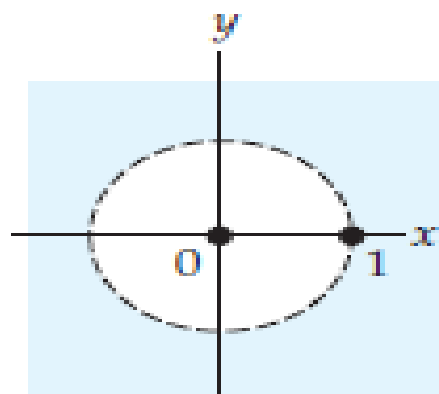
The infinite series in the brackets converges for $|z| < 1$, but after we multiply this expression by $1/z$, the resulting series

$$f(z) = -\frac{1}{z} - 1 - z - z^2 - z^3 - \cdots$$

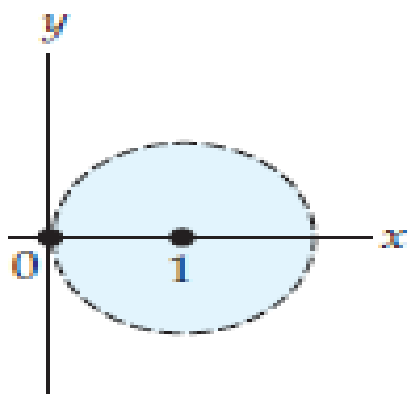
converges for $0 < |z| < 1$.



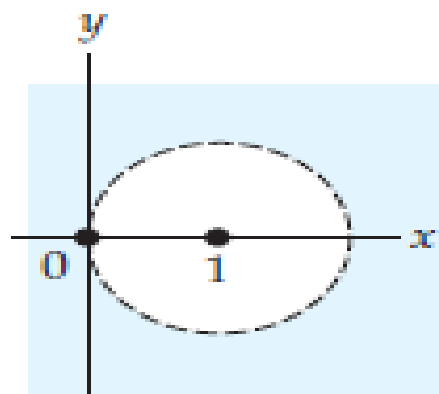
(a)



(b)



(c)



(d)

Annular domains for
Example 2

- (b) To obtain a series that converges for $1 < |z|$, we start by constructing a series that converges for $|1/z| < 1$. To this end we write the given function f as

$$f(z) = \frac{1}{z^2} \frac{1}{1 - \frac{1}{z}}$$

and $\frac{1}{1 - \frac{1}{z}}$ with z replaced by $1/z$:

$$f(z) = \frac{1}{z^2} \left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right].$$

The series in the brackets converges for $|1/z| < 1$ or equivalently for $1 < |z|$. Thus the required Laurent series is

$$f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \dots$$

- (c) This is basically the same problem as in part (a), except that we want all powers of $z - 1$. To that end, we add and subtract 1 in the denominator with z replaced by $z - 1$:

$$\begin{aligned} f(z) &= \frac{1}{(1 - 1 + z)(z - 1)} \\ &= \frac{1}{z - 1} \frac{1}{1 + (z - 1)} \\ &= \frac{1}{z - 1} [1 - (z - 1) + (z - 1)^2 - (z - 1)^3 + \dots] \\ &= \frac{1}{z - 1} - 1 + (z - 1) - (z - 1)^2 + \dots \end{aligned}$$

The requirement that $z \neq 1$ is equivalent to $0 < |z - 1|$, and the geometric series in brackets converges for $|z - 1| < 1$. Thus the last series converges for z satisfying $0 < |z - 1|$ and $|z - 1| < 1$, that is, for $0 < |z - 1| < 1$.

(d) Proceeding as in part (b), we write

$$\begin{aligned} f(z) &= \frac{1}{z-1} \frac{1}{1+(z-1)} = \frac{1}{(z-1)^2} \frac{1}{1+\frac{1}{z-1}} \\ &= \frac{1}{(z-1)^2} \left[1 - \frac{1}{z-1} + \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \dots \right] \\ &= \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \frac{1}{(z-1)^4} - \frac{1}{(z-1)^5} + \dots \end{aligned}$$

Because the series within the brackets converges for $|1/(z-1)| < 1$, the final series converges for $1 < |z-1|$.

EXAMPLE 3 Laurent Expansions

Expand $f(z) = \frac{1}{(z-1)^2(z-3)}$ in a Laurent series valid for (a) $0 < |z-1| < 2$ and (b) $0 < |z-3| < 2$.

Solution

(a) As in parts (c) and (d) of Example 2, we want only powers of $z-1$ and so we need to express $z-3$ in terms of $z-1$. This can be done by writing

$$f(z) = \frac{1}{(z-1)^2(z-3)} = \frac{1}{(z-1)^2} \frac{1}{-2+(z-1)} = \frac{-1}{2(z-1)^2} \frac{1}{1-\frac{z-1}{2}}$$

with the symbol z replaced by $(z-1)/2$,

$$\begin{aligned} f(z) &= \frac{-1}{2(z-1)^2} \left[1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \cdots \right] \\ &= -\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)} - \frac{1}{8} - \frac{1}{16}(z-1) - \cdots . \end{aligned}$$

(b) To obtain powers of $z - 3$, we write $z - 1 = 2 + (z - 3)$ and

We now factor 2
from this expression

$$\begin{aligned} f(z) &= \frac{1}{(z-1)^2(z-3)} = \frac{1}{z-3} \overbrace{[2 + (z-3)]}^{-2} \\ &= \frac{1}{4(z-3)} \left[1 + \frac{z-3}{2} \right]^{-2} \end{aligned}$$

At this point we can obtain a power series for $\left[1 + \frac{z-3}{2} \right]^{-2}$ by using the binomial expansion,[†]

$$f(z) = \frac{1}{4(z-3)} \left[1 + \frac{(-2)}{1!} \left(\frac{z-3}{2} \right) + \frac{(-2)(-3)}{2!} \left(\frac{z-3}{2} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(\frac{z-3}{2} \right)^3 + \dots \right].$$

The binomial series in the brackets is valid for $|(z-3)/2| < 1$ or $|z-3| < 2$. Multiplying this series by $\frac{1}{4(z-3)}$ gives a Laurent series that is valid for $0 < |z-3| < 2$:

$$f(z) = \frac{1}{4(z-3)} - \frac{1}{4} + \frac{3}{16}(z-3) - \frac{1}{8}(z-3)^2 + \dots.$$

[†]For α real, the binomial series $(1+z)^\alpha = 1 + \alpha z + \frac{\alpha(\alpha-1)}{2!}z^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}z^3 + \dots$ is valid for $|z| < 1$.

EXAMPLE 4 A Laurent Expansion

Expand $f(z) = \frac{8z + 1}{z(1 - z)}$ in a Laurent series valid for $0 < |z| < 1$.

Solution By partial fractions we can rewrite f as

$$f(z) = \frac{8z + 1}{z(1 - z)} = \frac{1}{z} + \frac{9}{1 - z}.$$

Then

$$\frac{9}{1 - z} = 9 + 9z + 9z^2 + \dots.$$

The foregoing geometric series converges for $|z| < 1$, but after we add the term $1/z$ to it, the resulting Laurent series

$$f(z) = \frac{1}{z} + 9 + 9z + 9z^2 + \dots$$

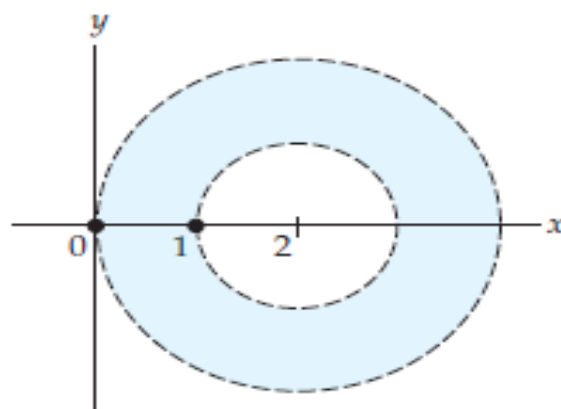
is valid for $0 < |z| < 1$.

EXAMPLE 5 A Laurent Expansion

Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z-2| < 2$.

Solution The specified annular domain is shown in Figure The center of this domain, $z = 2$, is the point of analyticity of the function f . Our goal now is to find two series involving integer powers of $z - 2$, one converging for $1 < |z - 2|$ and the other converging for $|z - 2| < 2$. To accomplish this, we proceed as in the last example by decomposing f into partial fractions:

$$f(z) = -\frac{1}{z} + \frac{1}{z-1} = f_1(z) + f_2(z). \quad ***$$



Annular domain for
Example 5

$$\begin{aligned}
\text{Now, } f_1(z) &= -\frac{1}{z} = -\frac{1}{2+z-2} \\
&= -\frac{1}{2} \frac{1}{1+\frac{z-2}{2}} \\
&= -\frac{1}{2} \left[1 - \frac{z-2}{2} + \frac{(z-2)^2}{2^2} - \frac{(z-2)^3}{2^3} + \dots \right] \\
&= -\frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \dots
\end{aligned}$$

This series converges for $|(z-2)/2| < 1$ or $|z-2| < 2$. Furthermore,

$$\begin{aligned}
f_2(z) &= \frac{1}{z-1} = \frac{1}{1+z-2} = \frac{1}{z-2} \frac{1}{1+\frac{1}{z-2}} \\
&= \frac{1}{z-2} \left[1 - \frac{1}{z-2} + \frac{1}{(z-2)^2} - \frac{1}{(z-2)^3} + \dots \right] \\
&= \frac{1}{z-2} - \frac{1}{(z-2)^2} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^4} + \dots
\end{aligned}$$

converges for $|1/(z-2)| < 1$ or $1 < |z-2|$. Substituting these two results in $\dots$ then gives

$$f(z) = \cdots - \frac{1}{(z-2)^4} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^2} + \frac{1}{z-2} - \frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \cdots$$

This representation is valid for z satisfying $|z-2| < 2$ and $1 < |z-2|$; in other words, for $1 < |z-2| < 2$.

EXAMPLE 6 A Laurent Expansion

Expand $f(z) = e^{3/z}$ in a Laurent series valid for $0 < |z| < \infty$.

Solution

we know that for all finite z , that is,

$|z| < \infty$,

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots . \quad *$$

We obtain the Laurent series for f by simply replacing z in $*$ by $3/z$, $z \neq 0$,

$$e^{3/z} = 1 + \frac{3}{z} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots . \quad **$$

This series $**$ is valid for $z \neq 0$, that is, for $0 < |z| < \infty$.

Find the Laurent series of $z^{-5} \sin z$ with center 0.

Solution.

$$z^{-5} \sin z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n-4} = \frac{1}{z^4} - \frac{1}{6z^2} + \frac{1}{120} - \frac{1}{5040} z^2 + \dots \quad (|z| > 0).$$

Here the "annulus" of convergence is the whole complex plane without the origin and the principal part of the series at 0 is $z^{-4} - \frac{1}{6} z^{-2}$.

Find the Laurent series of $z^2 e^{1/z}$ with center 0.

Solution.

If with z replaced by $1/z$ we obtain a Laurent series whose principal part is an infinite series,

$$z^2 e^{1/z} = z^2 \left(1 + \frac{1}{1z} + \frac{1}{2z^2} + \dots \right) = z^2 + z + \frac{1}{2} + \frac{1}{3z} + \frac{1}{4z^2} + \dots \quad (|z| > 0).$$

Develop $1/(1 - z)$ (a) in nonnegative powers of z , (b) in negative powers of z .

Solution.

$$(a) \quad \frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n \quad (\text{valid if } |z| < 1).$$

$$(b) \quad \frac{1}{1 - z} = \frac{-1}{z(1 - z^{-1})} = - \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} = -\frac{1}{z} - \frac{1}{z^2} - \dots \quad (\text{valid if } |z| > 1).$$

Find all Laurent series of $1/(z^3 - z^4)$ with center 0.

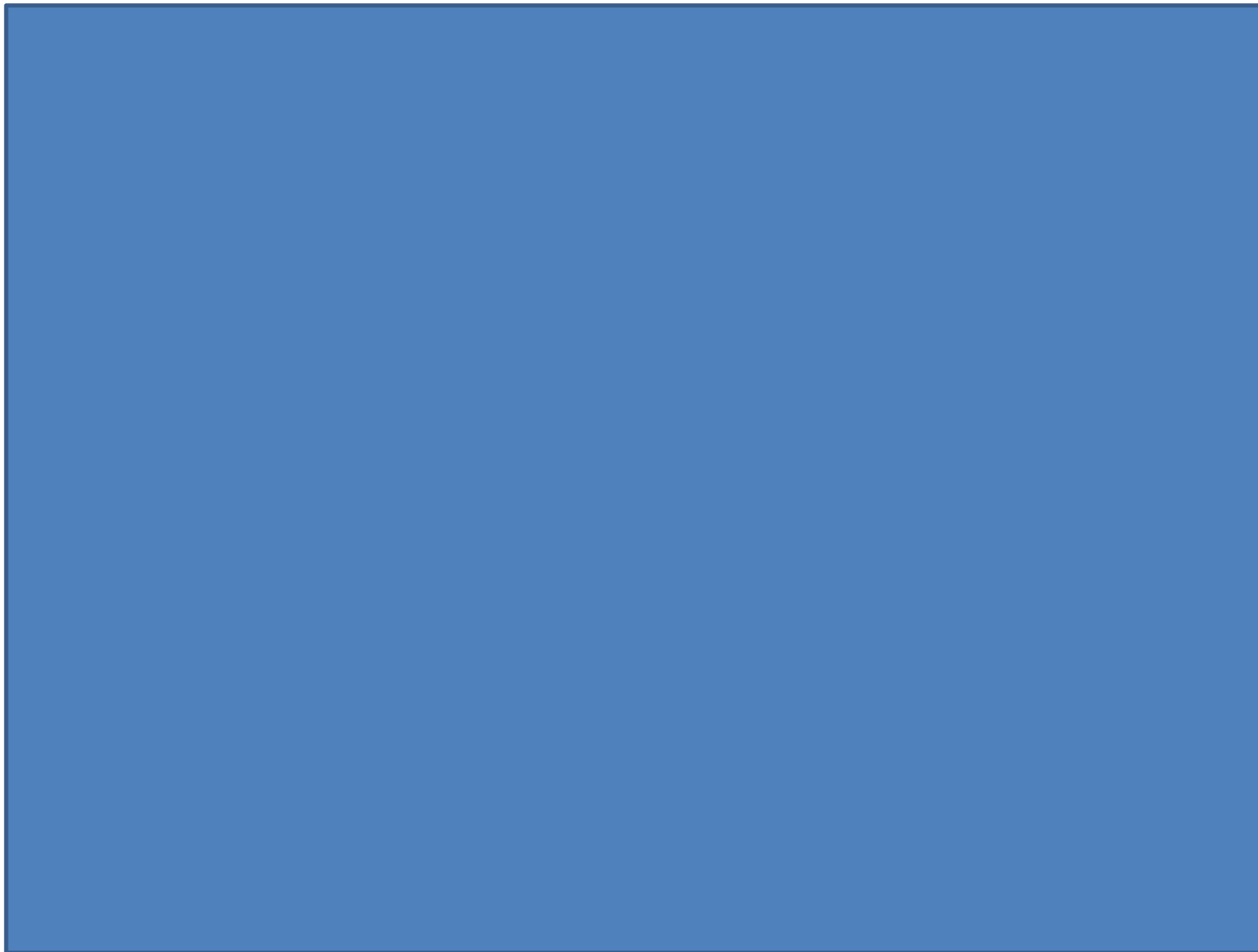
Solution. Multiplying by $1/z^3$, we get

$$(I) \quad \frac{1}{z^3 - z^4} = \sum_{n=0}^{\infty} z^{n-3} = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + \dots \quad (0 < |z| < 1),$$

$$(II) \quad \frac{1}{z^3 - z^4} = - \sum_{n=0}^{\infty} \frac{1}{z^{n+4}} = -\frac{1}{z^4} - \frac{1}{z^5} - \dots \quad (|z| > 1).$$

HOMEWORK

Find all Taylor and Laurent series of $f(z) = \frac{-2z + 3}{z^2 - 3z + 2}$ with center 0.





Zeros and Poles

Suppose $z = z_0$ is an isolated singularity of a complex function f , and that

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k = \sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z - z_0)^k \quad (1)$$

is the Laurent series representation of f valid for the punctured open disk $0 < |z - z_0| < R$. We saw in the preceding section that a Laurent series (1) consists of two parts. That part of the series in (1) with negative powers of $z - z_0$, namely,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \quad (2)$$

is the **principal part** of the series. In the discussion that follows we will assign different names to the isolated singularity $z = z_0$ according to the number of terms in the principal part.

Classification of Isolated Singular Points

An isolated singular point $z = z_0$ of a complex function f is given a classification depending on whether the principal part (2) of its Laurent expansion (1) contains zero, a finite number, or an infinite number of terms.

- (i) If the principal part is zero, that is, *all* the coefficients a_{-k} in (2) are zero, then $z = z_0$ is called a **removable singularity**.
- (ii) If the principal part contains a finite number of nonzero terms, then $z = z_0$ is called a **pole**. If, in this case, the last nonzero coefficient in (2) is a_{-n} , $n \geq 1$, then we say that $z = z_0$ is a **pole of order n** . If $z = z_0$ is pole of order 1, then the principal part (2) contains exactly one term with coefficient a_{-1} . A pole of order 1 is commonly called a **simple pole**.
- (iii) If the principal part (2) contains an infinitely many nonzero terms, then $z = z_0$ is called an **essential singularity**.

Table summarizes the form of a Laurent series for a function f when $z = z_0$ is one of the above types of isolated singularities. Of course, R in the table could be ∞ .

$z = z_0$	Laurent Series for $0 < z - z_0 < R$
Removable singularity	$a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots$
Pole of order n	$\frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-(n-1)}}{(z - z_0)^{n-1}} + \dots + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \dots$
Simple pole	$\frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots$
Essential singularity	$\dots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots$

Table Forms of Laurent series

EXAMPLE 1 Removable Singularity

by dividing the Maclaurin series for $\sin z$ by z , we see from

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \dots \quad (3)$$

that all the coefficients in the principal part of the Laurent series are zero. Hence $z = 0$ is a removable singularity of the function $f(z) = (\sin z)/z$.

EXAMPLE 2 Poles and Essential Singularity

(a) Dividing the terms of $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$ by z^2 shows that

$$\frac{\sin z}{z^2} = \overbrace{\frac{1}{z}}^{\text{principal part}} - \frac{z}{3!} + \frac{z^3}{5!} - \dots$$

for $0 < |z| < \infty$. From this series we see that $a_{-1} \neq 0$ and so $z = 0$ is a simple pole of the function $f(z) = (\sin z)/z^2$. In like manner, we see that $z = 0$ is a pole of order 3 of the function $f(z) = (\sin z)/z^4$.

- (b) we showed that the Laurent expansion of $f(z) = 1/(z-1)^2(z-3)$ valid for $0 < |z-1| < 2$ was

$$f(z) = \overbrace{-\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)}}^{\text{principal part}} - \frac{1}{8} - \frac{z-1}{16} - \cdots.$$

Since $a_{-2} = -\frac{1}{2} \neq 0$, we conclude that $z = 1$ is a pole of order 2.

- (c) the principal part of the Laurent expansion of the function $f(z) = e^{3/z}$ valid for $0 < |z| < \infty$ contains an infinite number of nonzero terms. This shows that $z = 0$ is an essential singularity of f .

Zeros

Recall, a number z_0 is **zero** of a function f if $f(z_0) = 0$. We say that an analytic function f has a **zero of order n** at $z = z_0$ if

z_0 is a zero of f and of its first $n-1$ derivatives

$$\overbrace{f(z_0) = 0, \ f'(z_0) = 0, \ f''(z_0) = 0, \ \dots, \ f^{(n-1)}(z_0) = 0} \text{, but } f^{(n)}(z_0) \neq 0. \quad (4)$$

A zero of order n is also referred to as a **zero of multiplicity n** . For example, for $f(z) = (z - 5)^3$ we see that $f(5) = 0$, $f'(5) = 0$, $f''(5) = 0$, but $f'''(5) = 6 \neq 0$. Thus f has a zero of order (or multiplicity) 3 at $z_0 = 5$. A zero of order 1 is called a **simple zero**.

The next theorem is a consequence of (4).

Theorem Zero of Order n

A function f that is analytic in some disk $|z - z_0| < R$ has a zero of order n at $z = z_0$ if and only if f can be written

$$f(z) = (z - z_0)^n \phi(z), \tag{5}$$

where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

EXAMPLE 3 Order of a Zero

The analytic function $f(z) = z \sin z^2$ has a zero at $z = 0$. If we replace z by z^2 in $\sin z$ we obtain the Maclaurin expansion

$$\sin z^2 = z^2 - \frac{z^6}{3!} + \frac{z^{10}}{5!} - \cdots.$$

Then by factoring z^2 out of the foregoing series we can rewrite f as

$$f(z) = z \sin z^2 = z^3 \phi(z) \quad \text{where} \quad \phi(z) = 1 - \frac{z^4}{3!} + \frac{z^8}{5!} - \cdots \quad (6)$$

and $\phi(0) = 1$. When compared to (5), the result in (6) shows that $z = 0$ is a zero of order 3 of f .

Theorem Pole of Order n

A function f analytic in a punctured disk $0 < |z - z_0| < R$ has a pole of order n at $z = z_0$ if and only if f can be written

$$f(z) = \frac{\phi(z)}{(z - z_0)^n}, \quad (7)$$

where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

Theorem Pole of Order n

If the functions g and h are analytic at $z = z_0$ and h has a zero of order n at $z = z_0$ and $g(z_0) \neq 0$, then the function $f(z) = g(z)/h(z)$ has a pole of order n at $z = z_0$.

Proof Because the function h has zero of order n , (5) gives $h(z) = (z - z_0)^n \phi(z)$, where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$. Thus f can be written

$$f(z) = \frac{g(z)/\phi(z)}{(z - z_0)^n}. \quad ****$$

Since g and ϕ are analytic at $z = z_0$ and $\phi(z_0) \neq 0$, it follows that the function g/ϕ is analytic at z_0 . Moreover, $g(z_0) \neq 0$ implies $g(z_0)/\phi(z_0) \neq 0$. We conclude from Theorem 1 that the function f has a pole of order n at z_0 .

When $n = 1$ in (5), we see that a zero of order 1, or a simple zero, in the denominator h of $f(z) = g(z)/h(z)$ corresponds to a simple pole of f .

EXAMPLE 4 Order of Poles

(a) Inspection of the rational function

$$f(z) = \frac{2z + 5}{(z - 1)(z + 5)(z - 2)^4}$$

shows that the denominator has zeros of order 1 at $z = 1$ and $z = -5$, and a zero of order 4 at $z = 2$. Since the numerator is not zero at any of these points, it follows from Theorem 1 and **** that f has simple poles at $z = 1$ and $z = -5$, and a pole of order 4 at $z = 2$.

(b) In Example 3 we saw that $z = 0$ is a zero of order 3 of $z \sin z^2$. From Theorem 1 we conclude that the reciprocal function $f(z) = 1/(z \sin z^2)$ has a pole of order 3 at $z = 0$.

We call $z = z_0$ an **isolated singularity** of $f(z)$ if $z = z_0$ has a neighborhood without further singularities of $f(z)$. *Example:* $\tan z$ has isolated singularities at $\pm\pi/2, \pm3\pi/2$, etc.; $\tan(1/z)$ has a nonisolated singularity at 0.

The function

$$f(z) = \frac{1}{z(z-2)^5} + \frac{3}{(z-2)^2}$$

has a simple pole at $z = 0$ and a pole of fifth order at $z = 2$. Examples of functions having an isolated essential singularity at $z = 0$ are

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n! z^n} = 1 + \frac{1}{z} + \frac{1}{2! z^2} + \dots$$

We saw in the last section that if a complex function f has an *isolated singularity* at a point z_0 , then f has a Laurent series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots ,$$

which converges for all z *near* z_0 . More precisely, the representation is valid in some deleted neighborhood of z_0 or punctured open disk $0 < |z - z_0| < R$. In this section our entire focus will be on the coefficient a_{-1} and its importance in the evaluation of contour integrals.

Residue The coefficient a_{-1} of $1/(z - z_0)$ in the Laurent series given above is called the **residue** of the function f at the isolated singularity z_0 . We shall use the notation

$$a_{-1} = \text{Res}(f(z), z_0)$$

to denote the residue of f at z_0 . Recall, if the principal part of the Laurent series valid for $0 < |z - z_0| < R$ contains a finite number of terms with a_{-n} the last nonzero coefficient, then z_0 is a pole of order n ; if the principal part of the series contains an infinite number of terms with nonzero coefficients, then z_0 is an essential singularity.

EXAMPLE 1 Residues

(a) In part (b) of Example 2 we saw that $z = 1$ is a pole of order two of the function $f(z) = \frac{1}{(z-1)^2(z-3)}$. From the Laurent series obtained in that example valid for the deleted neighborhood of $z = 1$ defined by $0 < |z - 1| < 2$,

$$f(z) = \frac{-1/2}{(z-1)^2} + \overbrace{\frac{-1/4}{z-1}}^{a_{-1}} - \frac{1}{8} - \frac{z-1}{16} - \dots$$

we see that the coefficient of $1/(z-1)$ is $a_{-1} = \text{Res}(f(z), 1) = -\frac{1}{4}$.

(b) we saw that $z = 0$ is an essential singularity of $f(z) = e^{3/z}$. Inspection of the Laurent series obtained in that example,

$$e^{3/z} = 1 + \overbrace{\frac{3}{z}}^{a_{-1}} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots,$$

$0 < |z| < \infty$, shows that the coefficient of $1/z$ is $a_{-1} = \text{Res}(f(z), 0) = 3$.

Theorem Residue at a Simple Pole

If f has a simple pole at $z = z_0$, then

$$\text{Res}(f(z), z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z). \quad (1)$$

Theorem Residue at a Pole of Order n

If f has a pole of order n at $z = z_0$, then

$$\text{Res}(f(z), z_0) = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}} (z - z_0)^n f(z). \quad (2)$$

Proof Because f is assumed to have pole of order n at $z = z_0$, its Laurent expansion convergent on a punctured disk $0 < |z - z_0| < R$ must have the form

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots ,$$

where $a_{-n} \neq 0$. We multiply the last expression by $(z - z_0)^n$,

$$(z - z_0)^n f(z) = a_{-n} + \cdots + a_{-2}(z - z_0)^{n-2} + a_{-1}(z - z_0)^{n-1} + a_0(z - z_0)^n + a_1(z - z_0)^{n+1} + \cdots$$

and then differentiate both sides of the equality $n - 1$ times:

$$\frac{d^{n-1}}{dz^{n-1}}(z - z_0)^n f(z) = (n - 1)!a_{-1} + n!a_0(z - z_0) + \cdots . \quad (3)$$

Since all the terms on the right-hand side after the first involve positive integer powers of $z - z_0$, the limit of (3) as $z \rightarrow z_0$ is

$$\lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}}(z - z_0)^n f(z) = (n - 1)!a_{-1}.$$

Solving the last equation for a_{-1} gives (2).

Notice that (2) reduces to (1) when $n = 1$.

EXAMPLE 2 Residue at a Pole

The function $f(z) = \frac{1}{(z-1)^2(z-3)}$ has a simple pole at $z = 3$ and a pole of order 2 at $z = 1$. Use Theorems [1](#) and [2](#) to find the residues.

Solution Since $z = 3$ is a simple pole, we use (1):

$$\operatorname{Res}(f(z), 3) = \lim_{z \rightarrow 3} (z-3)f(z) = \lim_{z \rightarrow 3} \frac{1}{(z-1)^2} = \frac{1}{4}.$$

Now at the pole of order 2, the result in (2) gives

$$\begin{aligned}\operatorname{Res}(f(z), 1) &= \frac{1}{1!} \lim_{z \rightarrow 1} \frac{d}{dz} (z-1)^2 f(z) \\ &= \lim_{z \rightarrow 1} \frac{d}{dz} \frac{1}{z-3} \\ &= \lim_{z \rightarrow 1} \frac{-1}{(z-3)^2} = -\frac{1}{4}.\end{aligned}$$

When f is not a rational function, calculating residues by means of (1) or (2) can sometimes be tedious. It is possible to devise alternative residue formulas. In particular, suppose a function f can be written as a quotient $f(z) = g(z)/h(z)$, where g and h are analytic at $z = z_0$. If $g(z_0) \neq 0$ and if the function h has a zero of order 1 at z_0 , then f has a simple pole at $z = z_0$ and

$$\operatorname{Res}(f(z), z_0) = \frac{g(z_0)}{h'(z_0)}. \quad (4)$$

To derive this result we shall use the definition of a zero of order 1, the definition of a derivative, and then (1). First, since the function h has a zero of order 1 at z_0 , we must have $h(z_0) = 0$ and $h'(z_0) \neq 0$. Second, by definition of the derivative

$$h'(z_0) = \lim_{z \rightarrow z_0} \frac{h(z) - \overbrace{h(z_0)}^0}{z - z_0} = \lim_{z \rightarrow z_0} \frac{h(z)}{z - z_0}.$$

We then combine the preceding two facts in the following manner in (1):

$$\text{Res}(f(z), z_0) = \lim_{z \rightarrow z_0} (z - z_0) \frac{g(z)}{h(z)} = \lim_{z \rightarrow z_0} \frac{g(z)}{\frac{h(z)}{z - z_0}} = \frac{g(z_0)}{h'(z_0)}.$$

EXAMPLE 3 Using (4) to Compute Residues

The polynomial $z^4 + 1$ can be factored as $(z - z_1)(z - z_2)(z - z_3)(z - z_4)$, where z_1, z_2, z_3 , and z_4 are the four distinct roots of the equation $z^4 + 1 = 0$

(or equivalently, the four fourth roots of -1). It follows from Theorem that the function

$$f(z) = \frac{1}{z^4 + 1}$$

has four simple poles.

Since $z_1 = e^{\pi i/4}$, $z_2 = e^{3\pi i/4}$, $z_3 = e^{5\pi i/4}$, and $z_4 = e^{7\pi i/4}$. To compute the residues, we use (4) of this section along with Euler's formula

$$\operatorname{Res}(f(z), z_1) = \frac{1}{4z_1^3} = \frac{1}{4}e^{-3\pi i/4} = -\frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z), z_2) = \frac{1}{4z_2^3} = \frac{1}{4}e^{-9\pi i/4} = \frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z), z_3) = \frac{1}{4z_3^3} = \frac{1}{4}e^{-15\pi i/4} = \frac{1}{4\sqrt{2}} + \frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z), z_4) = \frac{1}{4z_4^3} = \frac{1}{4}e^{-21\pi i/4} = -\frac{1}{4\sqrt{2}} + \frac{1}{4\sqrt{2}}i.$$

Of course, we could have calculated each of the residues in Example 3 using formula (1).

$$\begin{aligned}\operatorname{Res}(f(z), z_1) &= \lim_{z \rightarrow z_1} (z - z_1) \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)} \\ &= \frac{1}{(z_1 - z_2)(z_1 - z_3)(z_1 - z_4)} \\ &= \frac{1}{(e^{\pi i/4} - e^{3\pi i/4})(e^{\pi i/4} - e^{5\pi i/4})(e^{\pi i/4} - e^{7\pi i/4})}.\end{aligned}$$

Finally, we must do this process three more times.

$f(z) = (9z + i)/(z^3 + z)$ has a simple pole at i because $z^2 + 1 = (z + i)(z - i)$, and

$$\operatorname{Res}_{z=i} \frac{9z + i}{z(z^2 + 1)} = \lim_{z \rightarrow i} (z - i) \frac{9z + i}{z(z + i)(z - i)} = \left[\frac{9z + i}{z(z + i)} \right]_{z=i} = \frac{10i}{-2} = -5i.$$

$p(i) = 9i + i$ and $q'(z) = 3z^2 + 1$ we confirm the result,

$$\operatorname{Res}_{z=i} \frac{9z + i}{z(z^2 + 1)} = \left[\frac{9z + i}{3z^2 + 1} \right]_{z=i} = \frac{10i}{-2} = -5i.$$

$f(z) = 50z/(z^3 + 2z^2 - 7z + 4)$ has a pole of second order at $z = 1$ because the denominator equals $(z + 4)(z - 1)^2$ (verify).

$$\operatorname{Res}_{z=1} f(z) = \lim_{z \rightarrow 1} \frac{d}{dz} (z - 1)^2 f(z) = \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{50z}{z + 4} \right) = \frac{200}{5^2} = 8.$$

Theorem Cauchy's Residue Theorem

Let D be a simply connected domain and C a simple closed contour lying entirely within D . If a function f is analytic on and within C , except at a finite number of isolated singular points $z_1, z_2, \dots, z_n$ within C , then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f(z), z_k). \quad (5)$$

Integrate the function $f(z) = z^{-4} \sin z$ counterclockwise around the unit circle C .

Solution. we obtain the Laurent series

$$f(z) = \frac{\sin z}{z^4} = \frac{1}{z^3} - \frac{1}{3z} + \frac{1}{5} - \frac{z^3}{7} + \dots$$

which converges for $|z| > 0$ (that is, for all $z \neq 0$). This series shows that $f(z)$ has a pole of third order at $z = 0$ and the residue $b_1 = -\frac{1}{3}$.

$$\oint_C \frac{\sin z}{z^4} dz = 2\pi i b_1 = -\frac{\pi i}{3}.$$

Integrate $f(z) = 1/(z^3 - z^4)$ clockwise around the circle $C: |z| = \frac{1}{2}$.

Solution. $z^3 - z^4 = z^3(1 - z)$ shows that $f(z)$ is singular at $z = 0$ and $z = 1$. Now $z = 1$ lies outside C . Hence it is of no interest here. So we need the residue of $f(z)$ at 0. We find it from the Laurent series that converges for $0 < |z| < 1$.

$$\frac{1}{z^3 - z^4} = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + \dots \quad (0 < |z| < 1).$$

We see from it that this residue is 1. Clockwise integration thus yields

$$\oint_C \frac{dz}{z^3 - z^4} = -2\pi i \operatorname{Res}_{z=0} f(z) = -2\pi i.$$

EXAMPLE 4 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{1}{(z-1)^2(z-3)} dz$, where

- (a) the contour C is the rectangle defined by $x = 0$, $x = 4$, $y = -1$, $y = 1$,
- (b) and the contour C is the circle $|z| = 2$.

Solution

- (a) Since both $z = 1$ and $z = 3$ are poles within the rectangle we have from (5) that

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), 3)]$$

We found these residues in Example 2. Therefore,

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i \left[\left(-\frac{1}{4} \right) + \frac{1}{4} \right] = 0.$$

- (b) Since only the pole $z = 1$ lies within the circle $|z| = 2$, we have from (5)

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i \operatorname{Res}(f(z), 1) = 2\pi i \left(-\frac{1}{4} \right) = -\frac{\pi}{2}i.$$

EXAMPLE 5 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{2z + 6}{z^2 + 4} dz$, where the contour C is the circle $|z - i| = 2$.

Solution By factoring the denominator as $z^2 + 4 = (z - 2i)(z + 2i)$ we see that the integrand has simple poles at $-2i$ and $2i$. Because only $2i$ lies within the contour C , it follows from (5) that

$$\oint_C \frac{2z + 6}{z^2 + 4} dz = 2\pi i \operatorname{Res}(f(z), 2i).$$

But

$$\begin{aligned}\operatorname{Res}(f(z), 2i) &= \lim_{z \rightarrow 2i} (z - 2i) \frac{2z + 6}{(z - 2i)(z + 2i)} \\ &= \frac{6 + 4i}{4i} = \frac{3 + 2i}{2i}.\end{aligned}$$

Hence,

$$\oint_C \frac{2z + 6}{z^2 + 4} dz = 2\pi i \left(\frac{3 + 2i}{2i} \right) = \pi(3 + 2i).$$

EXAMPLE 6 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{e^z}{z^4 + 5z^3} dz$, where the contour C is the circle $|z| = 2$.

Solution Writing the denominator as $z^4 + 5z^3 = z^3(z + 5)$ reveals that the integrand $f(z)$ has a pole of order 3 at $z = 0$ and a simple pole at $z = -5$. But only the pole $z = 0$ lies within the given contour and so from (5) and (2) we have,

$$\begin{aligned}\oint_C \frac{e^z}{z^4 + 5z^3} dz &= 2\pi i \operatorname{Res}(f(z), 0) = 2\pi i \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 \cdot \frac{e^z}{z^3(z + 5)} \\ &= \pi i \lim_{z \rightarrow 0} \frac{(z^2 + 8z + 17)e^z}{(z + 5)^3} = \frac{17\pi}{125} i.\end{aligned}$$

EXAMPLE 7 Evaluation by the Residue Theorem

Evaluate $\oint_C \tan z \, dz$, where the contour C is the circle $|z| = 2$.

Solution The integrand $f(z) = \tan z = \sin z / \cos z$ has simple poles at the points where $\cos z = 0$. the only zeros $\cos z$

are the real numbers $z = (2n + 1)\pi/2$, $n = 0, \pm 1, \pm 2, \dots$. Since only $-\pi/2$ and $\pi/2$ are within the circle $|z| = 2$, we have

$$\oint_C \tan z \, dz = 2\pi i \left[\operatorname{Res} \left(f(z), -\frac{\pi}{2} \right) + \operatorname{Res} \left(f(z), \frac{\pi}{2} \right) \right].$$

With the identifications $g(z) = \sin z$, $h(z) = \cos z$, and $h'(z) = -\sin z$, we see from (4) that

$$\operatorname{Res} \left(f(z), -\frac{\pi}{2} \right) = \frac{\sin(-\pi/2)}{-\sin(-\pi/2)} = -1$$

and

$$\operatorname{Res} \left(f(z), \frac{\pi}{2} \right) = \frac{\sin(\pi/2)}{-\sin(\pi/2)} = -1.$$

Therefore,

$$\oint_C \tan z \, dz = 2\pi i[-1 - 1] = -4\pi i.$$

EXAMPLE 8 Evaluation by the Residue Theorem

Evaluate $\oint_C e^{3/z} dz$, where the contour C is the circle $|z| = 1$.

Solution As we have seen, $z = 0$ is an essential singularity of the integrand $f(z) = e^{3/z}$ and so neither formulas (1) and (2) are applicable to find the residue of f at that point. Nevertheless, we saw in Example 1 that the Laurent series of f at $z = 0$ gives $\text{Res}(f(z), 0) = 3$. Hence from (5) we have

$$\oint_C e^{3/z} dz = 2\pi i \text{Res}(f(z), 0) = 2\pi i (3) = 6\pi i.$$

Evaluate the following integral counterclockwise around any simple closed path such that (a) 0 and 1 are inside C , (b) 0 is inside, 1 outside, (c) 1 is inside, 0 outside, (d) 0 and 1 are outside.

$$\oint_C \frac{4 - 3z}{z^2 - z} dz$$

Solution. The integrand has simple poles at 0 and 1,

$$\operatorname{Res}_{z=0} \frac{4 - 3z}{z(z - 1)} = \left[\frac{4 - 3z}{z - 1} \right]_{z=0} = -4, \quad \operatorname{Res}_{z=1} \frac{4 - 3z}{z(z - 1)} = \left[\frac{4 - 3z}{z} \right]_{z=1} = 1.$$

Confirm this by (4). *Answer:* (a) $2\pi i(-4 + 1) = -6\pi i$, (b) $-8\pi i$, (c) $2\pi i$, (d) 0.

Integrate $(\tan z)/(z^2 - 1)$ counterclockwise around the circle $C: |z| = \frac{3}{2}$.

Solution. $\tan z$ is not analytic at $\pm\pi/2, \pm3\pi/2, \dots$, but all these points lie outside the contour C . Because of the denominator $z^2 - 1 = (z - 1)(z + 1)$ the given function has simple poles at ± 1 . We thus obtain

$$\begin{aligned}\oint_C \frac{\tan z}{z^2 - 1} dz &= 2\pi i \left(\operatorname{Res}_{z=1} \frac{\tan z}{z^2 - 1} + \operatorname{Res}_{z=-1} \frac{\tan z}{z^2 - 1} \right) \\ &= 2\pi i \left(\left. \frac{\tan z}{2z} \right|_{z=1} + \left. \frac{\tan z}{2z} \right|_{z=-1} \right) \\ &= 2\pi i \tan 1 = 9.7855i.\end{aligned}$$

Evaluate the following integral, where C is the ellipse $9x^2 + y^2 = 9$ (counterclockwise, sketch it).

$$\oint_C \left(\frac{ze^{\pi z}}{z^4 - 16} + ze^{\pi/z} \right) dz.$$

Solution. Since $z^4 - 16 = 0$ at $\pm 2i$ and ± 2 , the first term of the integrand has simple poles at $\pm 2i$ inside C , with residues note that $e^{2\pi i} = 1$

$$\operatorname{Res}_{z=2i} \frac{ze^{\pi z}}{z^4 - 16} = \left[\frac{ze^{\pi z}}{4z^3} \right]_{z=2i} = -\frac{1}{16},$$

$$\operatorname{Res}_{z=-2i} \frac{ze^{\pi z}}{z^4 - 16} = \left[\frac{ze^{\pi z}}{4z^3} \right]_{z=-2i} = -\frac{1}{16}$$

and simple poles at ± 2 , which lie outside C , so that they are of no interest here. The second term of the integrand has an essential singularity at 0, with residue $\pi^2/2$ as obtained from

$$ze^{\pi/z} = z \left(1 + \frac{\pi}{z} + \frac{\pi^2}{2z^2} + \frac{\pi^3}{3z^3} + \cdots \right) = z + \pi + \frac{\pi^2}{2} \cdot \frac{1}{z} + \cdots \quad (|z| > 0).$$

Answer: $2\pi i(-\frac{1}{16} - \frac{1}{16} + \frac{1}{2}\pi^2) = \pi(\pi^2 - \frac{1}{4})i = 30.221i$ by the residue theorem.

Some Consequences of the Residue Theorem

In this section we shall see how residue theory can be used to evaluate *real* integrals of the forms

$$\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta, \tag{1}$$

$$\int_{-\infty}^{\infty} f(x) dx, \tag{2}$$

$$\int_{-\infty}^{\infty} f(x) \cos \alpha x dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) \sin \alpha x dx, \tag{3}$$

where F in (1) and f in (2) and (3) are rational functions. For the rational function $f(x) = p(x)/q(x)$ in (2) and (3), we will assume that the polynomials p and q have no common factors.

In addition to evaluating the three integrals just given, we shall demonstrate how to use residues to evaluate real improper integrals that require integration along a branch cut.

The discussion ends with the relationship between the residue theory and the zeros of an analytic function and a consideration of how residues can, in certain cases, be used to find the sum of an infinite series.

Integrals of the Form $\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$

The basic idea here is to convert a real trigonometric integral of form (1) into a complex integral, where the contour C is the unit circle $|z| = 1$ centered at the origin.

To do this we begin with [Example 1](#) to parametrize this contour by $z = e^{i\theta}$, $0 \leq \theta \leq 2\pi$. We can then write

$$dz = ie^{i\theta} d\theta, \quad \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Since $dz = ie^{i\theta} d\theta = iz d\theta$ and $z^{-1} = 1/z = e^{-i\theta}$, these three quantities are equivalent to

$$d\theta = \frac{dz}{iz}, \quad \cos \theta = \frac{1}{2}(z + z^{-1}), \quad \sin \theta = \frac{1}{2i}(z - z^{-1}). \quad (4)$$

The conversion of the integral in (1) into a contour integral is accomplished by replacing, in turn, $d\theta$, $\cos \theta$, and $\sin \theta$ by the expressions in (4):

$$\oint_C F\left(\frac{1}{2}(z + z^{-1}), \frac{1}{2i}(z - z^{-1})\right) \frac{dz}{iz},$$

where C is the unit circle $|z| = 1$.

EXAMPLE 1 A Real Trigonometric Integral

Evaluate $\int_0^{2\pi} \frac{1}{(2 + \cos \theta)^2} d\theta$.

Solution When we use the substitutions given in (4), the given trigonometric integral becomes the contour integral

$$\oint_C \frac{1}{\left(2 + \frac{1}{2}(z + z^{-1})\right)^2} \frac{dz}{iz} = \oint_C \frac{1}{\left(2 + \frac{z^2 + 1}{2z}\right)^2} \frac{dz}{iz}.$$

Carrying out the algebraic simplification of the integrand then yields

$$\frac{4}{i} \oint_C \frac{z}{(z^2 + 4z + 1)^2} dz.$$

From the quadratic formula we can factor the polynomial $z^2 + 4z + 1$ as $z^2 + 4z + 1 = (z - z_1)(z - z_2)$, where $z_1 = -2 - \sqrt{3}$ and $z_2 = -2 + \sqrt{3}$. Thus, the integrand can be written

$$\frac{z}{(z^2 + 4z + 1)^2} = \frac{z}{(z - z_1)^2(z - z_2)^2}.$$

Because only z_2 is inside the unit circle C , we have

$$\oint_C \frac{z}{(z^2 + 4z + 1)^2} dz = 2\pi i \operatorname{Res}(f(z), z_2).$$

To calculate the residue, we first note that z_2 is a pole of order 2 and so we use (2)

$$\begin{aligned}\operatorname{Res}(f(z), z_2) &= \lim_{z \rightarrow z_2} \frac{d}{dz} (z - z_2)^2 f(z) = \lim_{z \rightarrow z_2} \frac{d}{dz} \frac{z}{(z - z_1)^2} \\ &= \lim_{z \rightarrow z_2} \frac{-z - z_1}{(z - z_1)^3} = \frac{1}{6\sqrt{3}}.\end{aligned}$$

Hence,
$$\frac{4}{i} \oint_C \frac{z}{(z^2 + 4z + 1)} dz = \frac{4}{i} \cdot 2\pi i \operatorname{Res}(f(z), z_1) = \frac{4}{i} \cdot 2\pi i \cdot \frac{1}{6\sqrt{3}}$$

and, finally,
$$\int_0^{2\pi} \frac{1}{(2 + \cos \theta)^2} d\theta = \frac{4\pi}{3\sqrt{3}}.$$

Show by the present method that $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = 2\pi$.

Solution. We use $\cos \theta = \frac{1}{2}(z + 1/z)$ and $d\theta = dz/iz$. Then the integral becomes

$$\begin{aligned} \oint_C \frac{dz/iz}{\sqrt{2} - \frac{1}{2}\left(z + \frac{1}{z}\right)} &= \oint_C \frac{dz}{-\frac{i}{2}(z^2 - 2\sqrt{2}z + 1)} \\ &= -\frac{2}{i} \oint_C \frac{dz}{(z - \sqrt{2} - 1)(z - \sqrt{2} + 1)}. \end{aligned}$$

We see that the integrand has a simple pole at $z_1 = \sqrt{2} + 1$ outside the unit circle C , so that it is of no interest here, and another simple pole at $z_2 = \sqrt{2} - 1$ (where $z - \sqrt{2} + 1 = 0$) inside C with residue

$$\begin{aligned} \operatorname{Res}_{z=z_2} \frac{1}{(z - \sqrt{2} - 1)(z - \sqrt{2} + 1)} &= \left[\frac{1}{z - \sqrt{2} - 1} \right]_{z=\sqrt{2}-1} \\ &= -\frac{1}{2}. \end{aligned}$$

Answer: $2\pi i(-2/i)(-\frac{1}{2}) = 2\pi$. (Here $-2/i$ is the factor in front of the last integral.)

